higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

PLANT AND EQUIPMENT
NQF LEVEL 3

19 NOVEMBER 2019

This marking guideline consists of 5 pages.

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QUESTION 1

1.1 A
1.2 C
1.3 B
1.4 D
1.5 E

(5 × 1) **[5]**

QUESTION 2

2.1 2.1.2
2.1.5
2.1.3
2.1.7
2.1.1
2.1.6
2.1.4
2.1.8

(8 × 1) (8)

2.2 2.2.1 Jack
- A portable device✓ that uses a mechanical or hydraulic lifting system to raise heavy objects to shorter heights✓

2.2.2 Safe working load
- The load that a machine✓ can safely lift, suspend or lower✓

2.2.3 Safety factor
- The unit of load, above the normal operating rating,✓ that a device can handle without failing✓

(3 × 2) (6)

2.3
- Stop.
- Move slowly.
- Extend the boom.
- Retract the boom.
- Hoist the load.
- Lower the boom and raise the load.
- Raise the boom and lower the load.
- Raise the boom.
- Finished with the crane.

(Any 6 × 1) (6)
[20]

QUESTION 3

3.1 3.1.1 Torque
3.1.2 Power
3.1.3 Work
(3 × 1) (3)

3.2
- They use less fuel than petrol engines doing the same work.
- They do not have an electrical ignition system.
- They generally last longer than petrol engines do.
- Diesel is a safer fuel than petrol is. (4 × 1) (4)

3.3
- Spark plugs
- Intake cam
- Exhaust cam
- Exhaust valves
- Clutch
- Gearbox
- Flywheel
- Crank
- Sump
- Connecting rod
- Pistons (Any 10 × 1) (10)

3.4 3.4.1 95%
3.4.2 10% and 30%
3.4.3 40%
(3 × 1) (3)
[20]

QUESTION 4

4.1 4.1.1 A torque converter is also a type of fluid coupling. (1)

4.1.2 It is used in motor cars with automatic transmissions✓ as well as in heavy earth-moving equipment.✓ (2 × 1) (2)

4.1.3 A torque converter is a gear reducer in equipment. (1)

4.1.4
- Heavy earth-moving equipment
- Dump trucks
- Bulldozers
- Rollers (Any 2 × 1) (2)

4.2
- It moves slowly.
- It is extremely heavy.
- It is very expensive to make. (Any 2 × 1) (2)

4.3
- Avoid clunking the transmission.
- Do not force gear shifts.
- Avoid misusing the clutch.
- Do not drive with your foot hovering on the clutch.
- Avoid wheel spins. (Any 2 × 1) (2)

[10]

QUESTION 5

5.1
- Locate the dipstick in the engine of a vehicle.
- Pull the dipstick out and wipe it with a clean cloth or paper towel.
- Insert the dipstick back into its slot and hold it there for a few seconds.
- Pull it out again and check the oil level that is displayed on the tip end of the dipstick.
- Wipe it clean again and push it back into its slot before topping up the oil, if needed. (5 × 1) (5)

5.2
- Brake warning light
- Engine oil warning light
- Petrol/fuel warning light
- Coolant level warning light
- Airbag warning light
- Battery warning light
- Engine warning light (Any 6 × 1) (6)

5.3 5.3.1 GVM
- Gross vehicle mass

5.3.2 GCM
- Gross combination mass

(2 × 1) (2)

5.4 5.4.1 Green light = fully charged/good battery
5.4.2 Yellow light = low battery
5.4.1 Red light = flat battery

(3 × 1) (3)

5.5
- Engine oil
- Oil filter
- Air filter
- Spark plugs (4)

[20]

QUESTION 6

6.1
- Air impact wrench
- Pneumatic grinder
- Pneumatic sander
- Pneumatic drill
- Pneumatic screwdriver
- Pneumatic nail gun (Any 5 × 1) (5)

6.2
- A sturdy and well-positioned main handle and front handle
- An accessible and easy-to-operate trigger
- Adjustment levers
- A line of sight
- Sawdust and chip ejection
- Exhaust airflow (Any 5 × 1) (5)

6.3
- Grinding disc
- Motor
- Handle
- Safety guard
- Electric cord (5)

[15]

QUESTION 7

7.1 7.1.1 Acetylene cylinders
They are stored in maroon cylinders✓ and are classified as low-pressure cylinders✓ in the industry. Oxy-fuel gases must always be used with the correct PPE and handled safely.✓

7.1.2 Oxygen cylinders
Oxygen is stored in taller, thinner, black cylinders and it supports combustion✓ and increases the flame temperature of any fuel gas which is burning.✓ Oxygen itself is not flammable.✓

(2 × 3) (6)

7.2
- Close the oxygen valve and then close the acetylene valve.
- Close the cylinder valves.
- Purge the hoses by opening the torch valves one at a time.
- Rest the regulators at zero outlet pressure and close both torch valves. (4)

[10]

TOTAL: 100

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